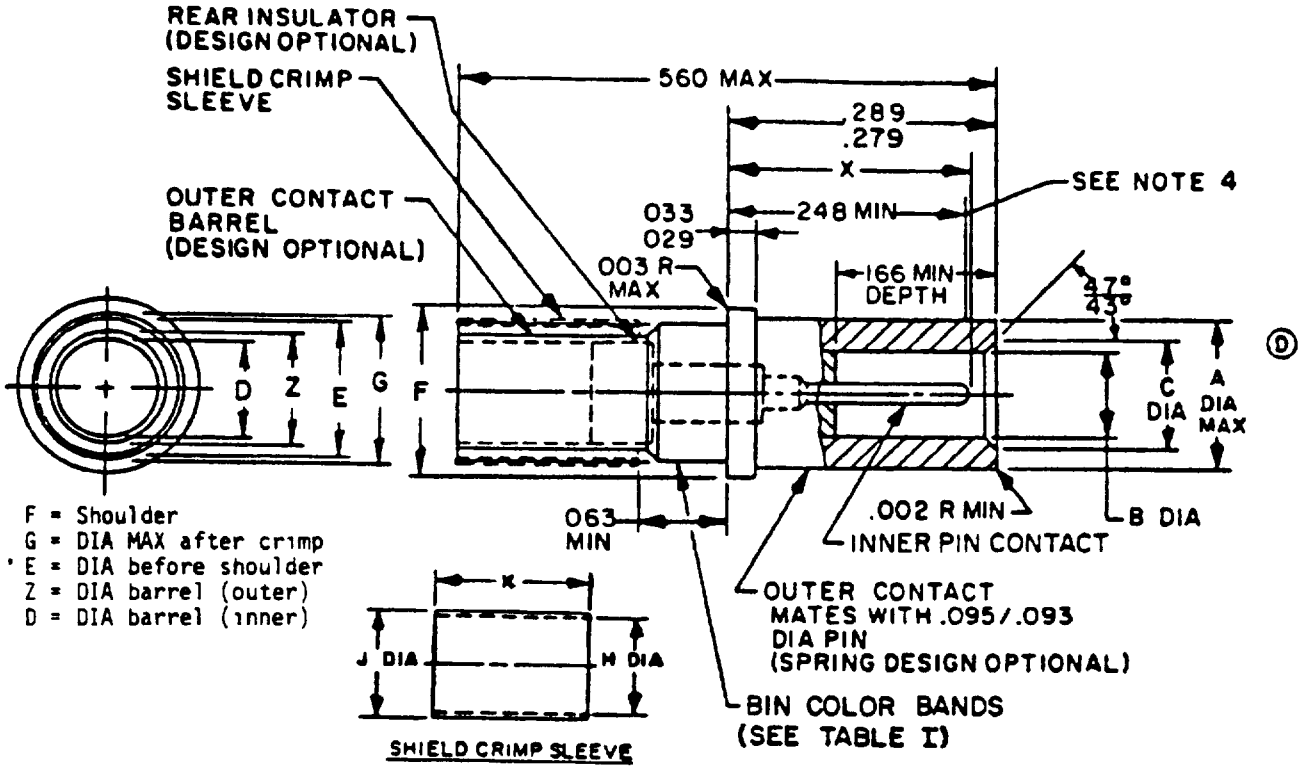
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, SIZE 12 (FOR MIL-C-38999 SERIES II CONNECTORS)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.



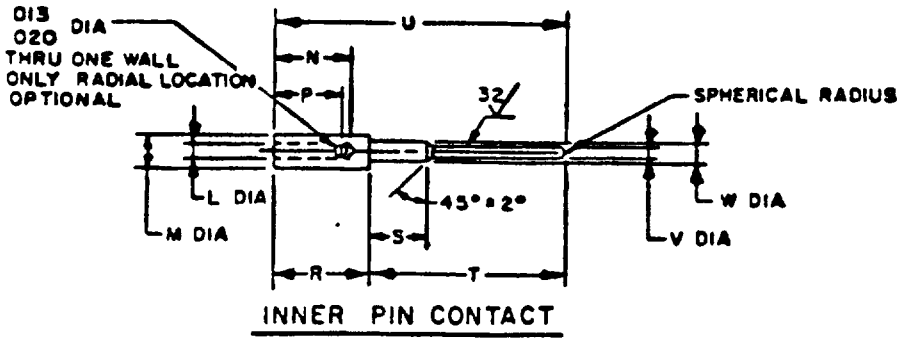


FIGURE 1. Socket contacts.

Ⓓ denotes changes

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(D)

BIN code	A Dia. MAX	B Dia.	C Dia.	D Dia. Min.	E Dia.	F Dia.	G Dia. Max.	H Dia. Min.	J Dia. Max.	K	L Dia. Min.	M Dia. Max.
210	.161	.100 .097	.123 .118	.090	.151 .148	.182 .179	.156	.127	.169	.125 .115	0225	.052
402				.108				.144			.0225	
403				.108				.144			.0355	
404				.090				.127			0355	
405				.090				.127	.169		.0225	
406				.117				.156	174		0270	
407				090				.144	.169		.0225	
408	.161	.100 .097	.123 .118	.108	.151 .148	.182 .179	.156	.156	.174	.125 115	.0355	.052

BIN code	N Min.	P	R	S	T	U REF.	V Dia.	W Dia.	X	Z Dia. Max.
210	.112	.103 .096	.146 .140	.039 .033	.222 219	.3635	.0205 .0195	.035 .033	.276 .266	.110
402										.127
403										.127
404										.110
405										.110
406										.136
407										110
408	.112	.103 .096	.146 .140	.039 .033	.222 219	.3635	.0205 .0195	.035 .033	276 .266	127

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	0.05	.033	0.84	.096	2.44	.118	3.00	.151	3.84	.219	5.56
.003	0.08	.035	0.89	.097	2.46	.123	3.12	.156	3.96	.222	5.64
.013	0.33	.0355	0.902	.100	2.54	.125	3.18	.158	4.01	.248	6.30
.0195	0.495	.039	0.99	.103	2.62	127	3.23	.161	4.09	.266	6.76
.020	0.51	.052	1.32	.108	2.74	.136	3.45	.166	4.22	.276	7.01
.0205	0.521	.063	1.60	.110	2.79	.140	3.56	.169	4.29	.279	7.09
.0225	0.572	.090	2.29	.112	2.84	.144	3.66	.174	4.42	.289	7.34
.027	0.69	.093	2.36	.115	2.92	.146	3.71	.179	4.55	.3635	9.233
.029	0.74	.095	2.41	.117	2.97	.148	3.76	.182	4.62	.560	14.22

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the outer contact spring. Provision for clearance hole shall be provided.
5. Crimp deformation: The maximum diameter over the crimped portion of the shield crimp sleeve shall not exceed G diameter.

FIGURE 1. Socket contacts - Continued.

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REQUIREMENTS.

Contacts shall comply with the reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

Dimensions, design characteristics, and configuration. See figure 1 and table I.

Mating contacts. MIL-C-39029/28.

Tools: See table II.

TABLE I. Design characteristics.

BIN code	Color bands			Cable accommodated	Contact cavity size	Type	Class
	1st	2nd	3rd				
210	Red	Brown	Black	M17/119-RG174 M17/113-RG316 M17/094-RG179 1/ 2/ Times AA3248 Teledyne 11299 Thermax 75-738-BCCWXE Tensolite 30888/L707YX-1 Haveg 8100207	12	D	B
402	Yellow	Black	Red	M17/095-RG180 Raychem 9527D1514-2L Raychem 9528A1318 1/ Microdot 293-3922	"	"	"
403	Yellow	Black	Orange	1/ Microdot 250-4070	"	"	"
404	Yellow	Black	Yellow	1/ Raychem 48-502 & 5022E5111	"	"	"
405	Yellow	Black	Green	1/ Raychem 48-950 & 9530D5117	"	"	"
406	Yellow	Black	Blue	1/ Raychem 7624D1311 Raychem 9527A1318	"	"	"
407	Yellow	Black	Violet	1/ Gore GWN1159A	"	"	"
408	Yellow	Black	Gray	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	"	"	"

1/ Or equivalent.

2/ High tensile strength copper alloy wire.

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TABLE II. Tools.

BIN code	Inner contact		Outer contact		Installing tool	Removal tool
	Basic crimping tool	Positioner	Basic crimping tool	Positioner		
210,402, 403,404, 405,406, 407,408	M22520/2-01	M22520/2-34	M22520/31-01	M22520/31-02	M81969/8-09 or M81969/14-04	M81969/8-10 or M81969/14-04

Mating cable. Mating cable shall be as specified in table III.

TABLE III. Mating cable to contact.

BIN code	Cable accommodated	Inner contact tool selector setting no.
210	M17/119-RG174	4
	M17/113-RG316	5
	M17/094-RG179	3
	Times AA3248	4
	Teledyne 11299	4
	1/ 2/ Thermax 75-738-BCCWXE	4
	Tensolite 30888/L707YX-1	4
	Haveg 8100207	4
402	M17/095-RG180	3
	Raychem 9527D1514-2L	5
	Raychem 9528A1318	4
	1/ Microdot 293-3922	3
403	1/ Microdot 250-4070	4
404	1/ Raychem 48-502 & 5022E5111	4
405	1/ Raychem 48-950 & 9530D5117	3
406	1/ Raychem 7624D1311	5
	Raychem 9527A1318	5
407	1/ Gore GWN1159A	4
408	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	4

1/ Or equivalent.

2/ High tensile strength copper alloy wire.

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Contact resistance. See table IV.

Test current·

Inner contact - 1 ampere.

Outer contact - 12 amperes.

Low signal level contact resistance (inner contact only): See table V.

Contact engagement and separation forces (socket contact only)· The engagement depth shall be as encountered in normal service. The test pins shall be in accordance with MS3197 except the diameters shall be as specified in the following, and surface roughness shall not exceed 3 microinches. Provision for clearance hole shall be provided.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.0950 +.0002 -.0000	NA	NA	30	36	NA
.0930 +.0000 -.0002	3.0	2.5	NA	NA	NA

Dielectric withstanding voltage (applied between inner and outer contact):

Test voltage:

At sea level - 1,000 V ac rms.

At 50,000 feet - 250 V ac rms.

Tensile strength (inner and outer contact crimp joint): See table V.

TABLE IV. Contact resistance.

BIN code	Cable accommodated	Maximum voltage drop (millivolts) 25°C +3°C -0°C Inner contact	25°C +3°C -0°C Outer contact	1/ 25°C +3°C -0°C Inner contact	25°C +3°C -0°C Outer contact	200°C +3°C -0°C Inner contact	200°C +3°C -0°C Outer contact	Maximum average voltage drop
210	M17/119-RG174	55	85	66	102	94 4/	145 4/	NA
	M17/113-RG316	55	75	66	90	94	128	"
	M17/094-RG179	120	70	144	84	204	119	"
	2/ 3/ Times AA3248	170	150	204	180	290	255	"
	Teledyne 11299	"	"	"	"	"	"	"
	Thermax 75-738-BCCWXE	"	"	"	"	"	"	"
	Tensolite 30888/L707YX-1	"	"	"	"	"	"	"
	Haveg 8100207	"	"	"	"	"	"	"

- - -

See footnotes at end of table.

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TABLE IV. Contact resistance - Continued.

BIN code	Cable accommodated	Maximum voltage drop (millivolts)						Maximum average voltage drop
		25°C +3°C -0°C		1/ 25°C +3°C -0°C		200°C +3°C -0°C		
		Inner contact	Outer contact	Inner contact	Outer contact	Inner contact	Outer contact	
402	M17/095-RG180	120	60	144	72	204	102	NA
	2/ Raychem 9527D1514-2L	"	"	"	"	204 5/	102 5/	"
	2/ Raychem 9528A1318	"	"	"	"	204 5/	102 5/	"
	2/ Microdot 293-3922	"	"	"	"	204	102	"
403	2/ Microdot 250-4070	"	"	"	"	204	102	"
404	2/ Raychem 48-502 & 5022E5111	55	"	66	"	94 5/	102 5/	"
405	2/ Raychem 48-950 & 9530D5117	120	"	144	"	204 5/	102 5/	"
406	2/ Raychem 7624D1311	55	"	66	"	94 5/	102 5/	"
	2/ Raychem 9527A1318	120	"	144	"	204 5/	102 5/	"
407	2/ Gore GWN1159A	120	"	144	"	204	102	"
408	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	55	"	66	"	94 4/	102 4/	"

1/ After conditioning.

2/ Or equivalent.

3/ High tensile strength copper alloy wire.

4/ 85°C +3°C -0°C.

5/ 115°C +3°C -0°C.

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TABLE V. Low signal level contact resistance (inner contact only) and tensile strength.

BIN code	Cable accommodated	Maximum contact resistance (milliohms) Initial	Maximum contact resistance (milliohms) After conditioning	Tensile load (pounds) (minimum) Inner contact	Tensile load (pounds) (minimum) Outer contact
210	M17/119-RG174	55	66	15.0	15.0
	M17/113-RG316	55	66	10.0	15.0
	M17/094-RG179	120	144	3.5	15.0
	1/ 2/ Times AA3248	170	204	7.0	35.0
	1/ 2/ Teledyne 11299	"	"	"	"
	1/ 2/ Thermax 75-738-BCCWXE	"	"	"	"
	1/ 2/ Tensolite 30888/L707YX-1	"	"	"	"
	1/ 2/ Haveg B100207	"	"	"	"
402	M17/095-RG180	120	144	3.5	20.0
	1/ Raychem 9527D1514-2L	"	"	6.0	"
	1/ Raychem 9528A1318	"	"	6.0	"
	1/ Microdot 293-3922	"	"	3.5	"
403	1/ Microdot 250-4070	"	"	4.0	"
404	1/ Raychem 48-502 & 5022E5111	55	66	10.0	"
405	1/ Raychem 48-950 & 9530D5117	120	144	3.5	"
406	1/ Raychem 7624D1311	55	66	6.0	"
	1/ Raychem 9527A1318	120	144	6.0	"
407	1/ Gore GWN1159A	120	144	4.0	"
408	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	55	66	15.0	"

1/ Or equivalent.

2/ High tensile strength copper alloy wire.

Vibration: MIL-STD-1344, method 2005, test condition VI, letter J, except directions shall be longitudinal and perpendicular. Time duration shall be 8 hours in each direction.

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Shipping: Figure 1 illustrates final assembly; parts are to be shipped unassembled in a suitable container. Manufacturers recommended assembly instructions shall be shipped with the unit package.

QPL evaluating activity Defense Electronics Supply Center (DESC-E), Dayton, OH 45444.

Military part number: See table VI.

TABLE VI. Military part number.

BIN code	Military part number	Supersedes
210	M39029/27-210	M39029/27-12A
402	M39029/27-402	M39029/27-12B
403	M39029/27-403	M39029/27-12C
404	M39029/27-404	M39029/27-12D
405	M39029/27-405	M39029/27-12E
406	M39029/27-406	M39029/27-12F
407	M39029/27-407	M39029/27-12G
408	M39029/27-408	M39029/27-12H

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities
Army - AR
Navy - EC, MC, OS
Air Force - 11, 15, 17, 99
DLA - ES

Agent.
DLA - ES

Preparing activity.
Air Force - 85

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